Stochastic Processes Semester 1

Extension Questions 2 – Solutions

1. (i) (a)

$$\begin{aligned}\mathbf{P}(S_{10} \geq 0) &= \sum_{k=4}^{10} \binom{10}{k} \left(\frac{2}{3}\right)^k \left(\frac{1}{3}\right)^{10-k} \\ &\approx 0.980.\end{aligned}$$

(b)

$$\begin{aligned}\mathbf{P}(S_{100} \geq 30) &\approx 1 - \Phi(-0.5657) \\ &\approx 0.714.\end{aligned}$$

(c)

$$\mathbf{P}(\text{there exists } n > 0 \text{ such that } S_n \leq -2) = \frac{1}{16}.$$

(ii) (a)

$$\begin{aligned}\mathbf{P}(S_{10} \geq 0) &= \sum_{k=5}^{10} \binom{10}{k} \left(\frac{3}{4}\right)^k \left(\frac{1}{4}\right)^{10-k} \\ &\approx 0.980.\end{aligned}$$

(b)

$$\begin{aligned}\mathbf{P}(S_{100} \geq 30) &\approx 1 - \Phi(-2.3094) \\ &\approx 0.990.\end{aligned}$$

(c)

$$\mathbf{P}(\text{there exists } n > 0 \text{ such that } S_n \leq -2) = \frac{1}{9}.$$

(iii) (a)

$$\begin{aligned}\mathbf{P}(S_{10} \geq 0) &= \sum_{k=6}^{10} \binom{10}{k} \left(\frac{1}{3}\right)^k \left(\frac{2}{3}\right)^{10-k} \\ &\approx 0.077.\end{aligned}$$

(b)

$$\begin{aligned}\mathbf{P}(S_{100} \geq 30) &\approx 1 - \Phi(6.9296) \\ &\approx 2.109 \times 10^{-12}\end{aligned}$$

(c)

$$\mathbf{P}(\text{there exists } n > 0 \text{ such that } S_n \leq -2) = 1.$$

(iv) (a)

$$\mathbf{P}(S_{10} \geq 0) = \sum_{k=7}^{10} \binom{10}{k} \left(\frac{1}{2}\right)^{10}$$
$$\approx 0.172.$$

(b)

$$\mathbf{P}(S_{100} \geq 30) \approx 1 - \Phi(3.4)$$
$$\approx 3.369 \times 10^{-4}$$

(c)

$$\mathbf{P}(\text{there exists } n > 0 \text{ such that } S_n \leq -2) = 1.$$

2.

$$p = \frac{3}{16}$$

3.

$$p = \frac{1}{6}$$
$$q = \frac{1}{12}$$

4. (i) The probability this company ever runs out of capital is **at least** $\left(\frac{6}{7}\right)^3 = 0.630 > 0.513 = \left(\frac{7}{8}\right)^5$

(ii) (a)

$$\pi_{20} \approx 0.435$$

(b)

$$s.d. \approx 0.035$$

(c) You would require $n \geq 12989$

5. The basic game is a simple random walk with $p = q = 1/2$.

We are interested in the return.

We make a renumbering of boxes: box k becomes box $k-1$, for $k = 1, \ldots, 10$. Now we have boxes $0, 1, \ldots, 9$ and start from box $a = 4$.

We will solve a more general problem, with any sequence of boxes $0, 1, \ldots, b-1, b$, with any initial position a where $0 \leq a \leq b$, and with arbitrary cost of a game c (in place of 0.1) and return K (in place of 6).

Let R_a be the amount Bob receives eventually. We like to find $r_a = \mathbf{E}R_a$. Clearly, R_a is a random variable which may take only two values, K and 0, but the corresponding probabilities are unknown, apart from the boundary cases:
if $a = 0$, then $R_0 = 0$ with probability 1 and, therefore, $r_0 = \mathbf{E}R_0 = 0$; similarly, if $a = b$, then $R_b = K$ with probability 1 and, therefore, $r_b = \mathbf{E}R_b = K$.

Now, for any $0 < a < b$,

$$\mathbf{E}R_a = -c + \frac{1}{2}\mathbf{E}R_{a+1} + \frac{1}{2}\mathbf{E}R_{a-1}$$

where c is the cost of a game. So,

$$r_a = -c + \frac{1}{2}r_{a+1} + \frac{1}{2}r_{a-1}, \tag{1}$$

and we have to solve the system of linear equations (1) with boundary conditions $r_0 = 0$ and $r_b = K$.

We multiply both sides by 2 and rearrange the terms as follows:

$$r_{a+1} - r_a = 2c + r_a - r_{a-1}.$$

Let $r_a - r_{a-1} = d_a$ and let $d_1 = d = r_1 - r_0 = r_1$. Then $d_2 = 2c + d$, $d_3 = 2c + d_2 = 4c + d$, etc. By induction,

$$d_a = 2(a-1)c + d, \quad \text{for any} \quad a = 1, 2, \ldots, b.$$

Clearly,

$$r_a = r_a - r_0 = \sum_1^a d_i = da + 2c(1 + 2 + \ldots + a - 1) = da + ca(a-1). \tag{2}$$

In particular, for $a = b$,

$$K = r_b = db + cb(b-1),$$

and we may find

$$d = \frac{K}{b} - c(b-1).$$

Finally, from (2), we obtain a general formula: for any a,

$$r_a = \left(\frac{K}{b} - c(b-1)\right) a + ca(a-1).$$

Substituting the particular values $K = 6$, $b = 9$, $a = 4$ and $c = 0.1$, we get

$$r_4 = (\frac{6}{9} - 0.8) \cdot 4 + 0.1 \cdot 4 \cdot 3 = \frac{2}{3}.$$